One-dimensional Kinematics

Kinematics is the mathematical description of motion without forces

Note on notation

- Write equations for a generic "s-axis"
- Position: s
- Velocity: v_s
- Acceleration: a_s
- In actual problem, use variable of axis, e.g., x or y

Uniform motion

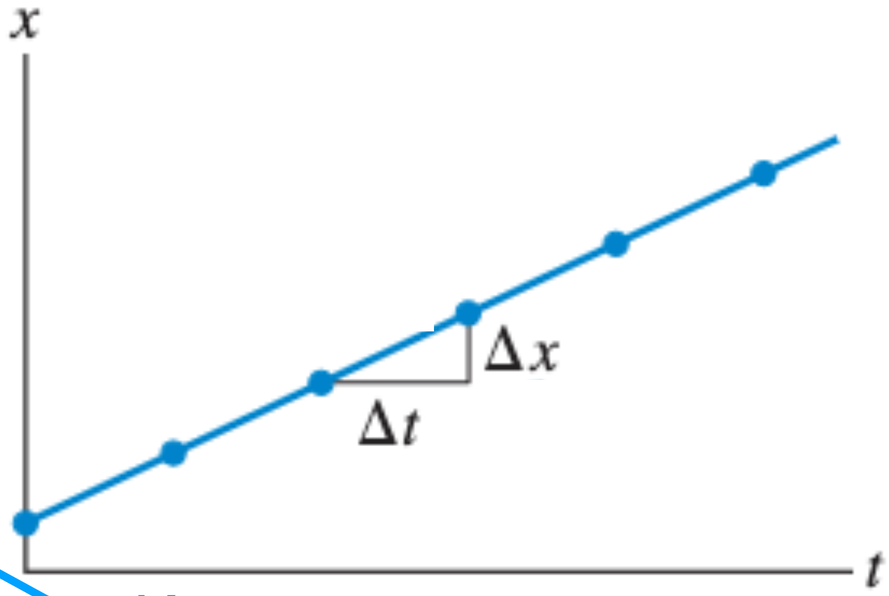
Motion diagram: uniform motion



- Simplest possible 1D motion
- Motion along a straight line at constant velocity
- Position vs. time graph is a straight line
- Because constant velocity:

$$v_s = \frac{\text{rise}}{\text{run}} = \frac{\Delta s}{\Delta t} = \frac{s_\text{f} - s_\text{i}}{t_\text{f} - t_\text{i}}$$

Position vs. time graph: uniform motion



Uniform motion model

Know Δs, Δt

Know s_i, v_s

Rearrange: $s_\text{f} = s_\text{i} + v_s \underbrace{\Delta t}_{t_\text{f} - t_\text{i}}$

Example

- Bob leaves home in Chicago at 9:00 AM and drives east at 60 mph. Susan, 400 miles to the east in Pittsburg, leaves at the same time and travels west at 40 mph. Where will they meet for lunch?

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- $\Delta t = 4.0$ hours
- 240 miles

Instantaneous velocity

Motion diagram of non-uniform motion

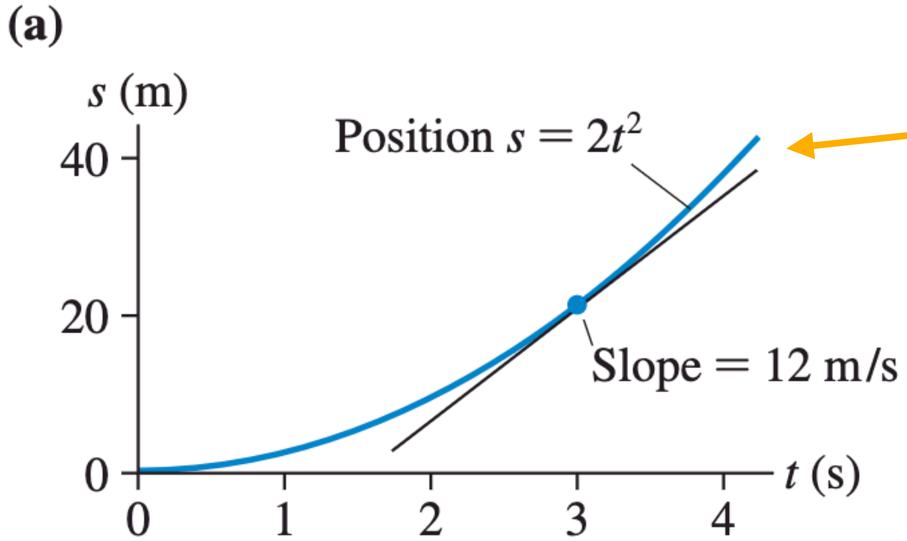


Velocity vector length increases, the position vs. time graph curves upward

- Consider non-uniform motion
- Average velocity better approximates true velocity the smaller the time interval (calculus)

$$v_s \equiv \lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt}$$

- Instantaneous velocity is:
  - Velocity **at a single instant of time**
  - the derivative of s with respect to t
  - the slope of the line tangent to the position-vs.-time graph at time t



Note about derivatives

- Most common functions we will use in Parts I and II of the book are powers and polynomials.

- Given $u = c =$ constant, what is $\dfrac{\mathrm{d}u}{\mathrm{d}t}$?

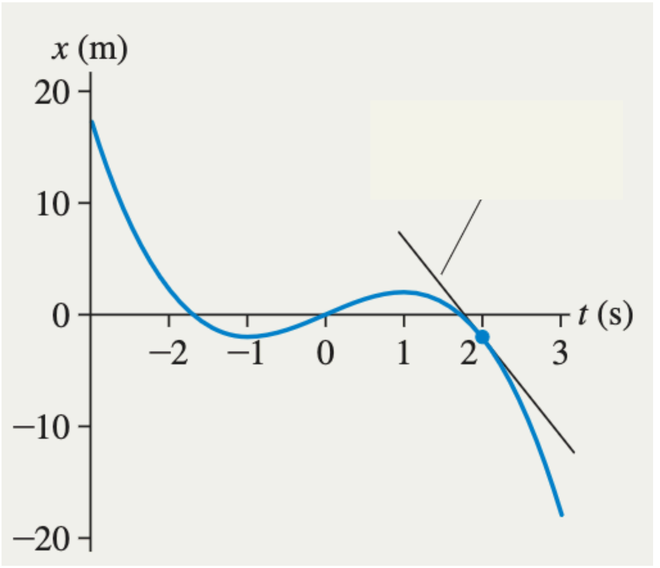
- Given $u = ct^n$, what is $\dfrac{\mathrm{d}u}{\mathrm{d}t}$?

- $\dfrac{\mathrm{d}}{dt}(u + w)$

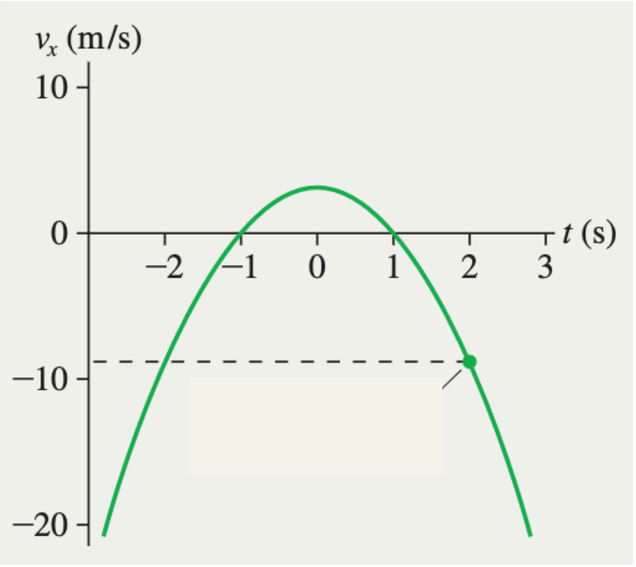
Example

- A particle's position is $x(t) = (-t^3 + 3t)\,\mathrm{m}$, what is the position and velocity at $t + 2\,\mathrm{s}$?

Position vs. time



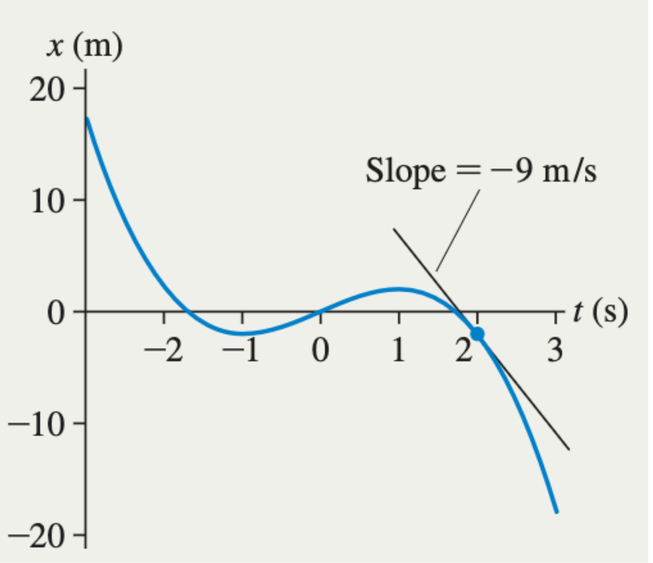
Velocity vs. time



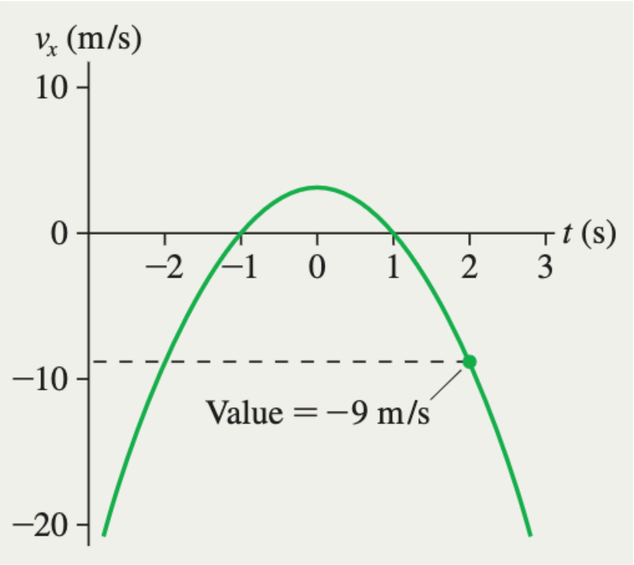
Example

- A particle's position is $x(t) = (-t^3 + 3t)\,\mathrm{m}$, what is the position and velocity at $t + 2\,\mathrm{s}$?
- $v_x = (-3t^2 + 3)\,\mathrm{m/s}$
- $v_x(2) = -9\,\mathrm{m/s}$

Position vs. time

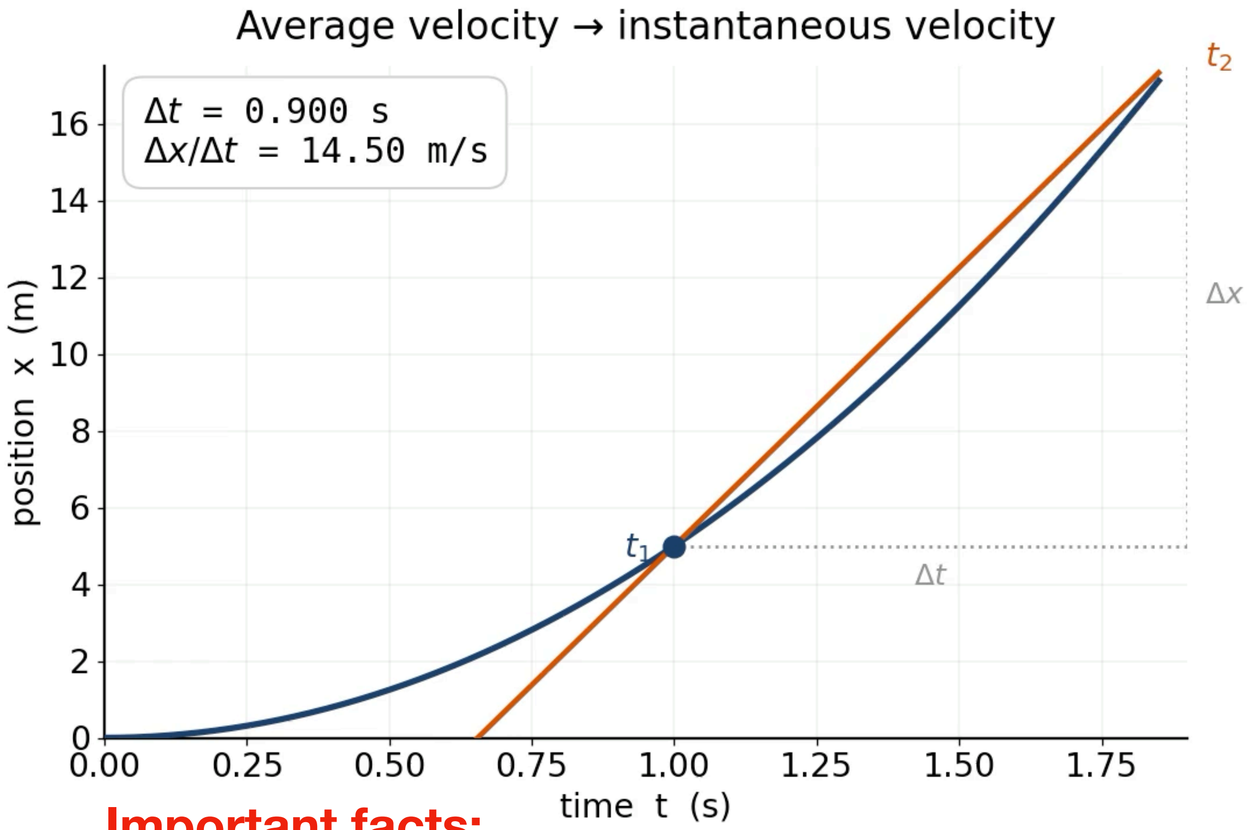


Velocity vs. time



Average vs. Instantaneous Velocity

- Average velocity
  - slope = straight line through two points on position curve
  - $v_{\text{avg}} = \Delta x / \Delta t$ (“*rise over run*”)
  - Think of as constant velocity need to take you from x_1 to x_2 in given Δt
  - Tip summary (e.g., 300km in 3 hours)
- Instantaneous velocity
  - slope of curve itself at one t
  - What speedometer of car reports
  - Quantity physics cares about; equations you will use later use instantaneous velocity

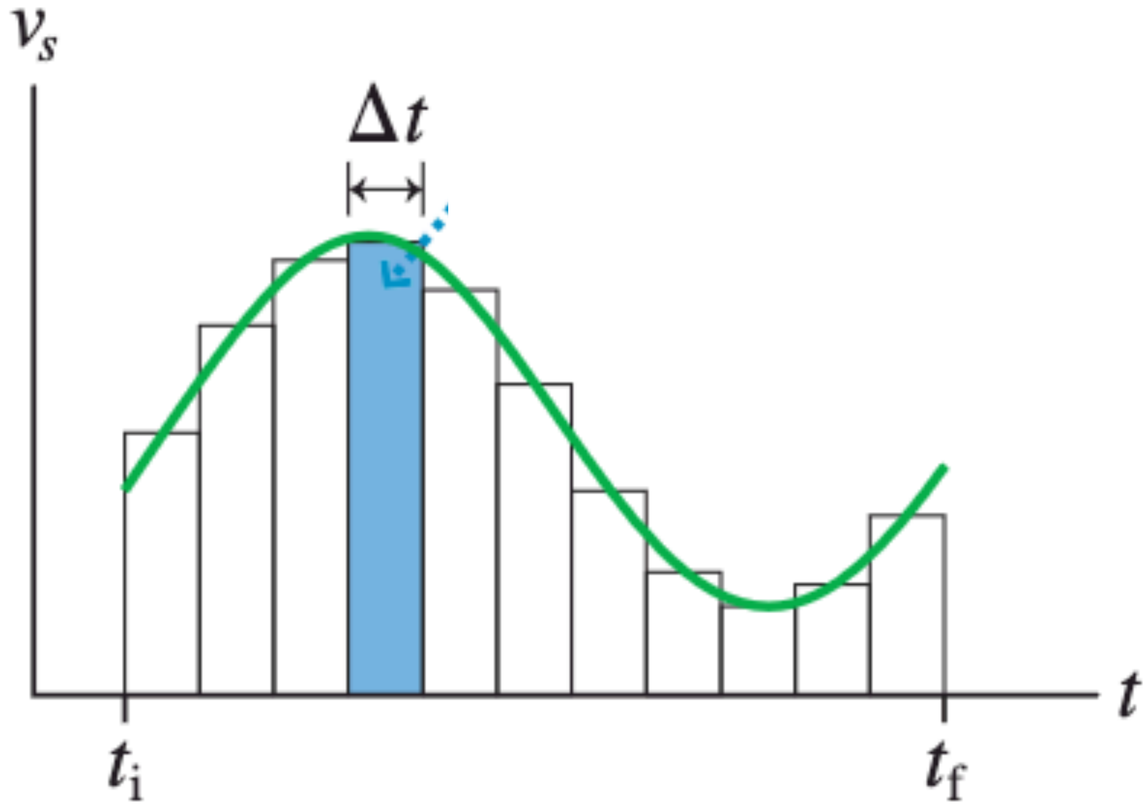


Important facts:

- Both velocities are slopes of position-vs.-time graph
- Curve of position-vs.-time graph is straight line = **constant velocity** (uniform motion).
- Curve of velocity-vs.-time graph is a straight line = **Constant acceleration**.

Position from velocity

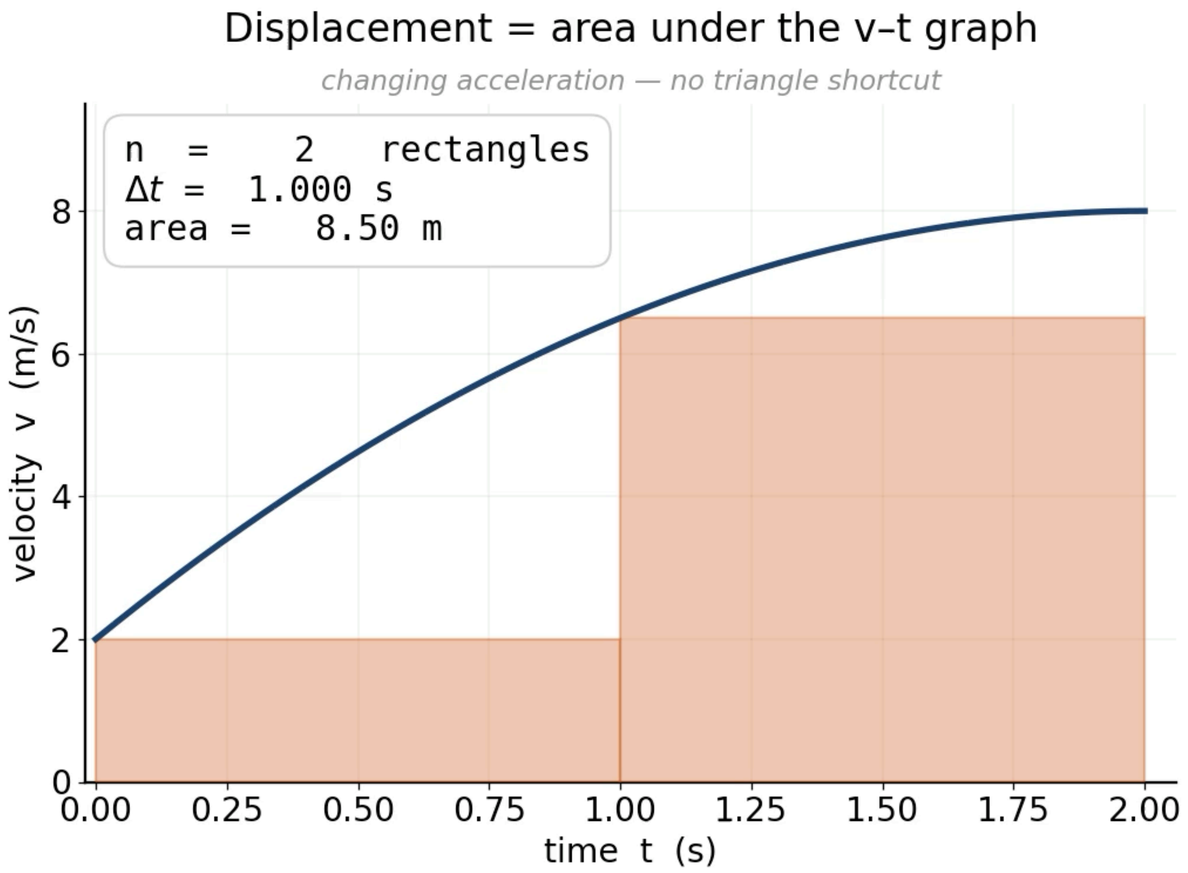
- Recall velocity is from position (slope of position-vs.-time graph)
- Now want total displacement Δs from velocity-vs.-time graph
- Displacement = area under the velocity-vs.-time graph
  - Divide area under curve into rectangles of width Δt
  - **Assume** velocity of each rectangle is **constant**.
  - Add up area from all rectangles



Position from velocity

- Area rectangle = width × height = $(v_s)_k \Delta t$
  - Last equality from average velocity
- $$\Delta s = s_\mathrm{f} - s_\mathrm{i} \approx \sum_{k_i}^{N} (v_s)_k \Delta t$$
- As $\Delta t \to 0$ series approaches exact solution
- $$s_\mathrm{f} = s_\mathrm{i} + \lim_{\Delta t \to 0} \sum_{k=1}^{N} (v_s)_k \Delta t = s_\mathrm{i} + \int_{t_\mathrm{i}}^{t_\mathrm{f}} v_s dt$$

t_1 and t_2 are the *upper* and *lower* bounds and make the integral a *definite integral*



$\sum$ (uppercase Greek sigma) symbol for summation

A note about integrals

Don’t get too caught up by calculus.

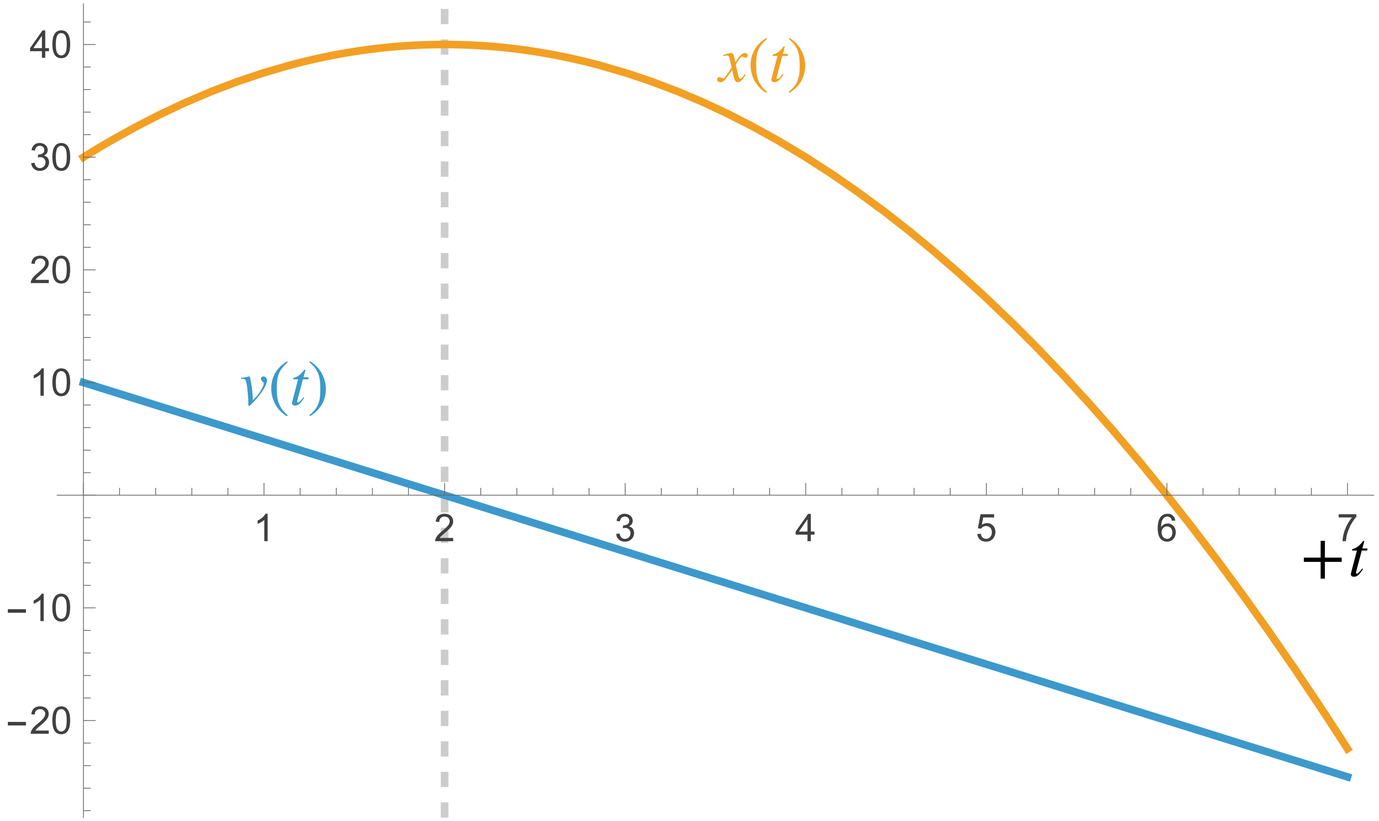
We will use it a little but not everything.

The stuff you need I’m giving you as a recipe.

- Integrals = area under curve
- Integrals “undo” derivatives (fundamental rule of calculus)
- What you need to be able to do:
  - $u(t) = ct^2$
  - $\int_{t_i}^{t_f} (u + w)\, dt$
- Vertical bar short hand notation = “integral evaluated at t_f minus evaluated at t_i”

Example

- Given $v_x = (10\text{m/s} - 5t)$, find $x(t)$ where $x_i = 30\,\text{m}$, $t_i = 0\,\text{s}$
- Where is the particle's turning point (i.e., where the position graph changes directions)?
- At what point does the particle reach the origin (i.e., where $x(t) = 0$)?



- Find $x(t)$ where $x_i = 30\,\mathrm{m}$, $t_i = 0\,\mathrm{s}$

$x = 30\mathrm{m} + 10\,t - 5/2\,t^2$

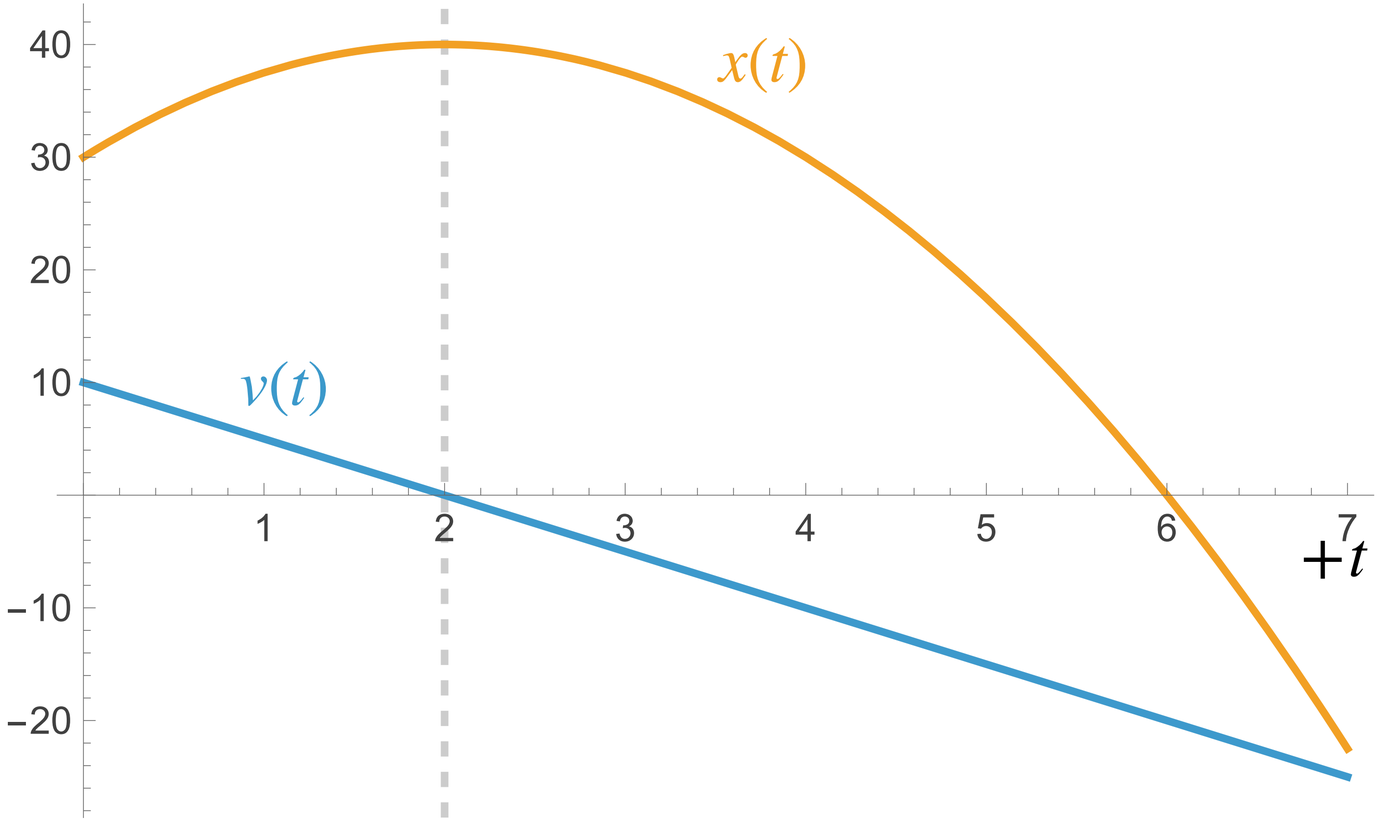
- Where is the particle's turning point?

$t = 2\,\mathrm{s}$

- At what point does the particle reach the origin?

$30\,\mathrm{m} + 10\,t - 5/2\,t^2 = 0 \rightarrow (-2,6)$

Take 6



Motion with constant acceleration

- An object whose velocity-vs.-time graph is a straight-line has constant acceleration
- When acceleration is constant $a_{\text{avg}} = \dfrac{\Delta v_s}{\Delta t} = a_{\text{inst}}$
- Acceleration is the change of velocity in the change in time $(\text{m/s})/\text{s}$
  - For example $(5\text{m/s})/\text{s}$
- SI units of acceleration: m/s^2

Kinematics equations (constant acceleration)

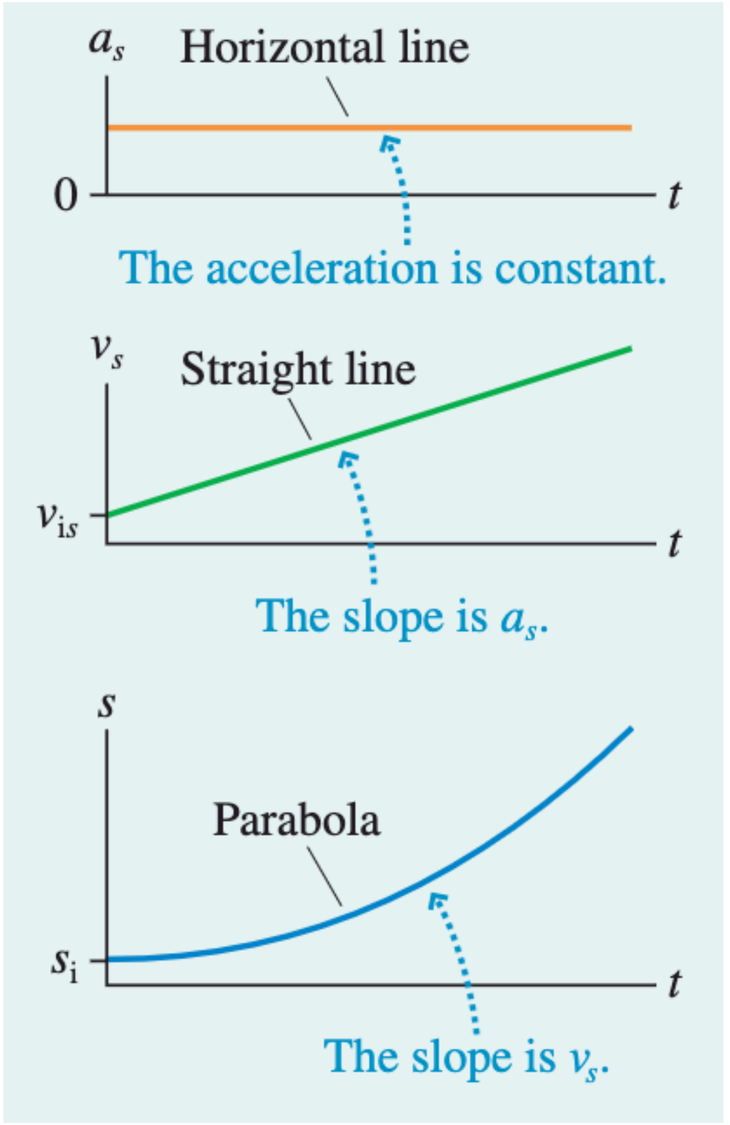
- **Average acceleration equation:** $a_s = \frac{\Delta v_s}{\Delta t} = \frac{v_{f,s} - v_{i,s}}{\Delta t}$
- $v_{f,s} = v_{i,s} + a_s \Delta t$
- $s_f = s_i + v_{is}\Delta t + \frac{1}{2} a_s (\Delta t)^2$

Kinematics equations (constant acceleration)

- Let $\Delta t = (v_{fs} - v_{is})/a_s$
- $v_{fs}^2 = v_{is}^2 + 2a_s\Delta s$
- $\Delta s = s_f - s_i$ is the displacement (not the distance)

Constant acceleration model

- Assume object has constant acceleration
- $v_{fs} = v_{is} + a_s \Delta t$
- $s_f = s_i + v_{is}\Delta t + \frac{1}{2} a_s \Delta t^2$
- $v_{fs}^2 = v_{is}^2 + 2a_s \Delta s$
- **Model fails if particle acceleration changes**



Announcement

- Let me know which HW question is the hardest/easiest and why.

Example

- A rocket sled's engines fire for $5.0\,\mathrm{s}$, boosting the sled to a speed of $250\,\mathrm{m/s}$. The sled then deploys a breaking parachute, slowing by $3.0\,\mathrm{m/s^2}$ until it stops. What is the total distance traveled?

Physics Problem Solving Tips

- Draw a diagram (if possible)
- Write down what you know
- Write down what you want to find
- First solve symbolically
- Check to make sure your units are correct
- If you end up with a number, check order of magnitude

1D Kinematic Equations

- $v_{\mathrm{f}s} = v_{\mathrm{i}s} + a_s \Delta t$
- $s_{\mathrm{f}} = s_{\mathrm{i}} + v_{\mathrm{i}s}\Delta t + \frac{1}{2} a_s \Delta t^2$
- $v_{\mathrm{f}s}^2 = v_{\mathrm{i}s}^2 + 2 a_s \Delta s$

Example

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- $a_{0x} = \dfrac{v_{1x}}{t_1} = \dfrac{250\,\mathrm{m/s}}{5.0\,\mathrm{s}} = 50\,\mathrm{m/s}$
- $x_1 = 625\,\mathrm{m}$ (accelerating)
- $x_2 = 11{,}000\,\mathrm{m}$ (full distance)

- Fred is driving his VW Beetle at a steady $20\,\mathrm{m/s}$ when he passes Betty setting at rest in her Porsche. Betty instantly begins begins accelerating at $5\,\mathrm{m/s^2}$. How far does Betty have to drive to overtake Fred?

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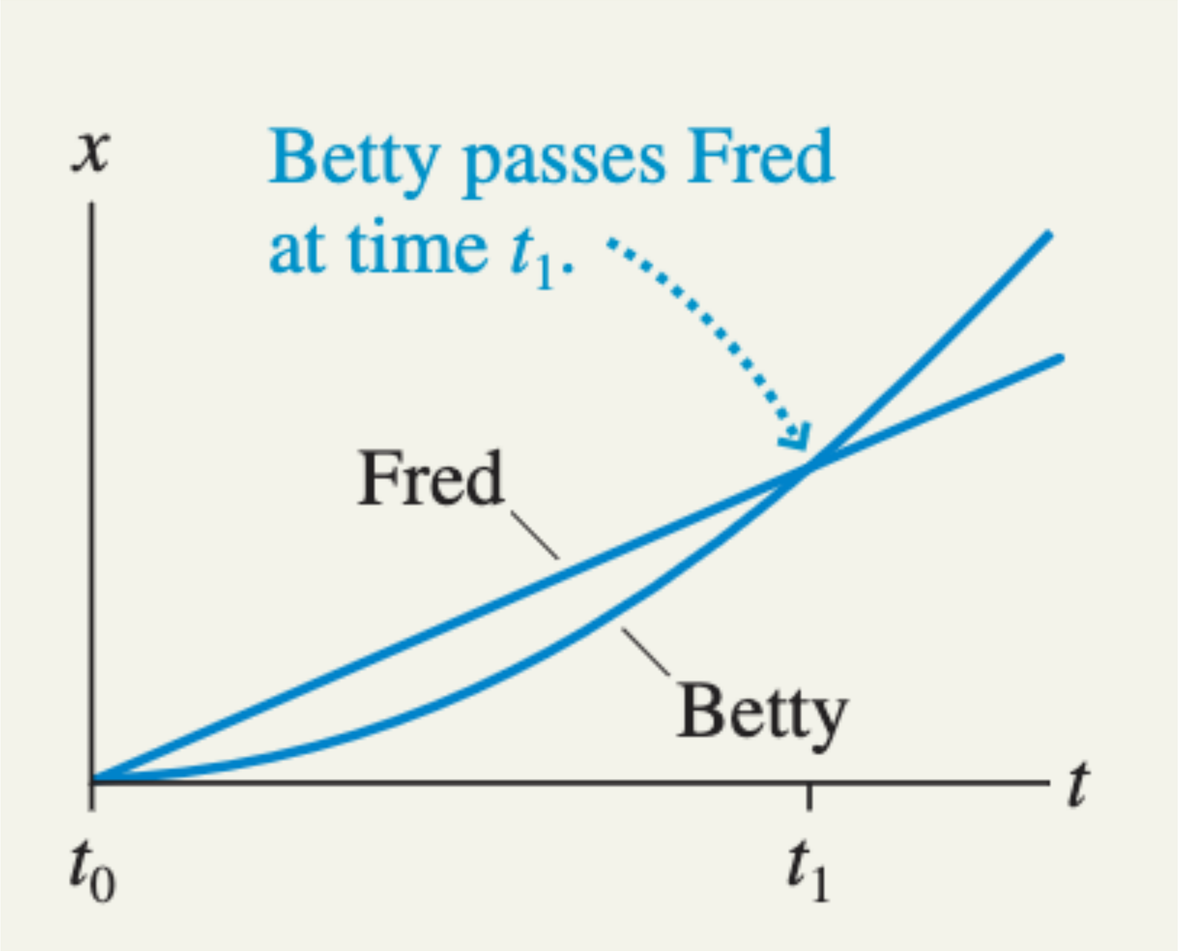
- $v_{\mathrm{f}s} = v_{\mathrm{i}s} + a_s \Delta t$
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- Fred is driving his Volkswagen Beetle at a steady $20\,\mathrm{m/s}$ when he passes Betty setting at rest in her Porsche. Betty instantly begins begins accelerating at $5\,\mathrm{m/s^2}$. How far does Betty have to drive to overtake Fred?

- $t_1 = 8.0\,\mathrm{s}$

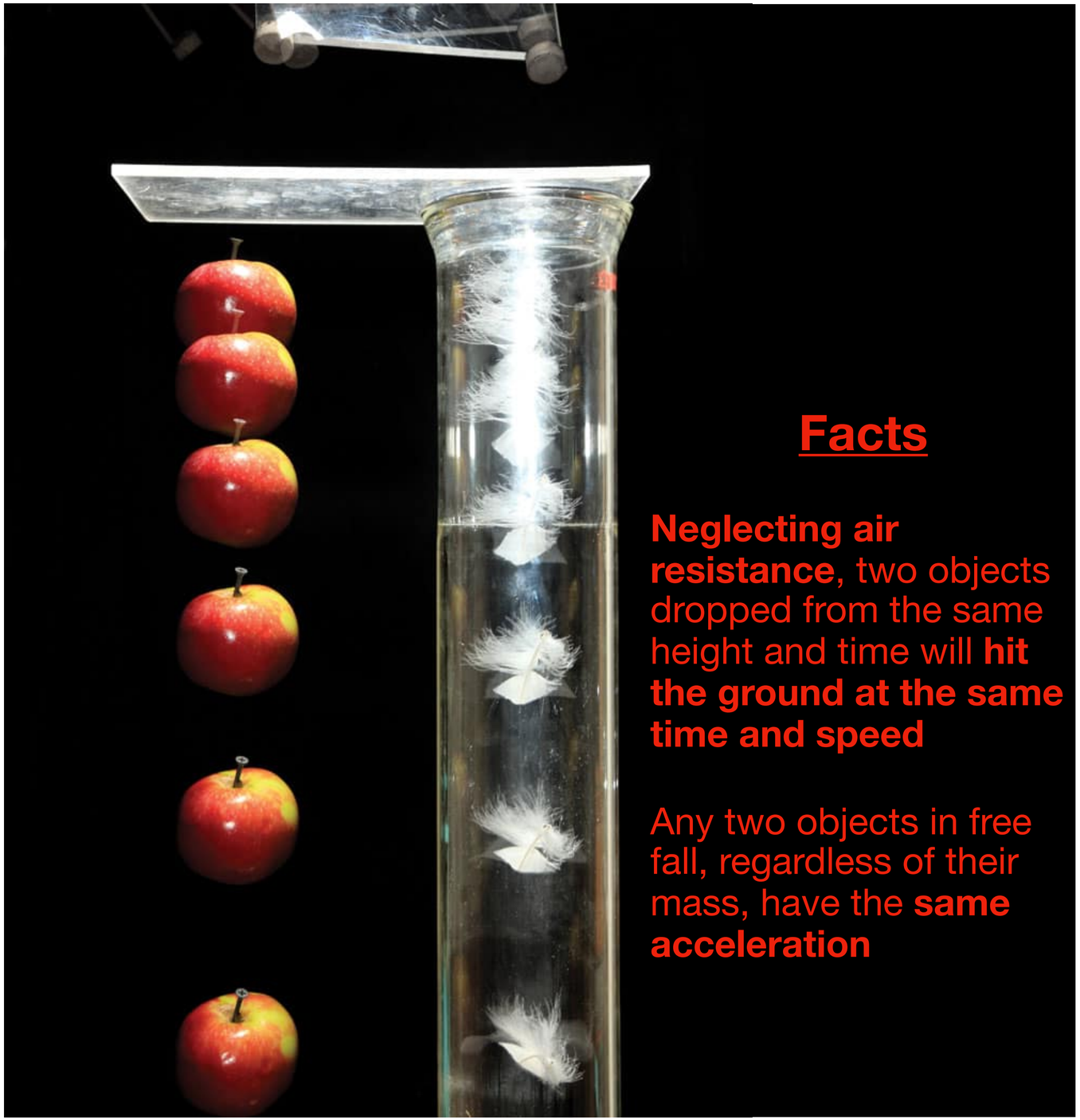
- Two solutions: when Fred first passes Betty ($t_0 = 0\,\mathrm{s}$) and $t_1 = 8.0\,\mathrm{s}$ when Betty passes Fred.

- $x_1 = 160\,\mathrm{m}$



Free Fall

- Motion of object under the influence of gravity **only (a**ssume **no air resistance)**
- $\vec{a}_{\text{free fall}}$
  - points towards the center of the earth
  - Magnitude (on earth) $g = 9.80\text{m/s}^2$ (other planets have different values for g)
  - g is “the free-fall acceleration” or “the acceleration due to gravity” *(***not gravity**; gravity is a force)
- Not just objects literally falling; **any object moving under the influence of gravity only**



Facts

Neglecting air resistance, two objects dropped from the same height and time will **hit the ground at the same time and speed**

Any two objects in free fall, regardless of their mass, have the **same acceleration**

Example

- A rock is dropped from the top of a $20\,\mathrm{m}$ tall building. What is its impact velocity?

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- $v = 20\mathrm{m/s}$

Motion on an Incline Plane

- Free fall motion, but different
- Assume **frictionless, incline plane**
- $\vec{a}_{\text{free fall}}$ still points down
- Solution steps
  - $\vec{a}_{\text{free fall}}$ broken into two pieces:
    - parallel with surface $a_{\parallel}$
    - perpendicular with surface $a_{\perp}$
  - Solve 1D motion with $\vec{a}_{\parallel}$

